Hallam Land Management, Taylor Wimpey & Hannick Homes

Eastern Villages, Swindon

REPTILE SURVEY REPORT

July 2013

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1.0 INTRODUCTION

1.1 This report has been prepared by FPCR Environment and Design Ltd on behalf of a consortium comprising Hallam Land Management Ltd, Taylor Wimpey and Hannick homes to provide details details of survey conducted for reptiles within and around a site to the east of Swindon. This report presents the results of survey and mitigation strategy to prevent adverse impacts on this species.

Site Context

1.2 The Study Area is centred on the Ordnance Survey Grid Reference SU 2007 8752 and lies to the east of Swindon close to the village of South Marston.

1.3 The site is primarily used for agriculture at present, with pastoral fields bisected by hedgerow field boundaries in the west, to the south of South Marston, and a range of agricultural uses centred around Rowborough Farm to the east, including arable use, turf production and biomass fuel production.

1.4 The site and wider area to the south has been the focus of a wide range of surveys from at least 1999, including a number completed by FPCR. These surveys included reptile surveys over a number of years both within the site and wider Eastern Villages Allocated site area.

2.0 LEGISLATION AND POLICY

2.1 All common reptile species, including common lizard and grass snake, are partially protected under Sections 9(1) and 9(5) of Schedule 5 of the Wildlife and Countryside Act 1981 (as amended). This legislation protects these animals from:

- intentional killing and injury;
- selling, offering for sale, possessing or transporting for the purpose of sale or publishing advertisements to buy or sell a protected species.

2.2 The impact that this legislation has on the Planning system is outlined in ODPM 06/2005 Government Circular: Biodiversity and Geological Conservation ï Statutory obligations and their Impact within the Planning System. This states:

The presence of a protected species is a material consideration when a planning authority is considering a development proposal that, if carried out, would be likely to result in harm to the species or its habitat. Local authorities should consult English Nature [now Natural England] *before granting planning permission. They should consider attaching appropriate planning conditions or entering into planning obligations under which the developer would take steps to secure the long-term protection of the species. They should also advise developers that they must comply with any statutory species' protection provisions affecting the site concerned.*

2.3 This partial protection does not directly protect the habitat of these reptile species. Where these animals are present on land that is to be affected by development, the implications of legislation are that providing that killing can reasonably be avoided then an operation is legal. Guidance

provided by Natural England[1] and the Amphibian and Reptile Groups of the UK[2] recommends that this should be achieved by ensuring that:

- the animals are protected from injury or killing;
- mitigation is provided to maintain the conservation status of the species;
- population monitoring is carried out subsequent to operations.

2.4 All common reptiles are also included on the list of habitats and species which are of principal importance for the conservation of biodiversity in England as required under Section 41 of the Natural Environment and Rural Communities (NERC) Act. The S41 list is used to guide decision-makers, including local planning authorities, in implementing their duty under section 40 of the Act, to have regard to the conservation of biodiversity in England, when carrying out their normal functions.

3.0 METHODOLOGY

Desktop study

3.1 In order to compile existing baseline information, relevant ecological information was searched for on online databases and requested from both statutory and non-statutory nature conservation organisations including:

- Wiltshire and Swindon Biological Records Centre (WSBRC
- Multi Agency Geographic Information for the Countryside (MAGIC) website (http://magic.defra.gov.uk)
- National Biodiversity Network (NBN) website (http://www.searchnbn.net)

Reptile Survey

3.2 A strategic reptile presence / absence survey was undertaken at specific locations identified as offering potential habitat within the area of survey.

3.3 The survey was undertaken based on methodology detailed in the *Herpetofauna Workers Manual*[3] and *the Froglife Advice Sheet 10 - Reptile Survey*[4]. Methods involved a search for basking reptiles on / under naturally occurring and strategically positioned artificial refugia (81 no). The artificial refugia used were 0.5m^2 sections of roofing felt with a black upper side. These were placed in locations that offered the most suitable habitat for common reptiles i.e. on the ephemeral margins between the hard standing and woodland and in open grassland and at a density of five refugia per hectare of suitable habitat. The location of refugia is shown in Figure 1.

3.4 In addition, the surveys also followed the guidelines recommendations by:

[1] Reptiles: guidelines for developers, English Nature (2004). http://publications.naturalengland.org.uk/publication/76006?category=31018

[2] Maintaining best practise in reptile mitigation/translocation programmes: Herpetofauna Groups of Britain and Ireland. http://www.arguk.org/index.php?option=com_docman&task=cat_view&gid=13&Itemid=17

[3] Herpetofauna Workers Manual, Gent and Gibson (1999). JNCC

[4] Froglife Advice Sheet 10: An introduction to planning, conducting and interpreting surveys for snake and lizard conservation http://www.froglife.org/documents/FroglifeAdviceSheet10.pdf

- Approaching refugia from downwind and avoiding casting a shadow and with care so as to not disturb basking animals when checking;
- Lifting and replacing tins, to check for the presence of reptiles underneath in hot weather is undertaken with care, to avoid potential harm to any animals underneath;

3.5 All of the surveys were undertaken between September and mid-October 2012 by suitably experienced FPCR ecologists. The surveys were carried out in appropriate weather conditions i.e. air temperature between 10-20°C, no strong wind or heavy rain (Table 1).

Table 1: Reptile Survey Weather Conditions

Survey occasion	Date	Start time	End time	Cloud cover	Max temp	Min temp	wind	rain
1	31/05/2013	08:00	11:00	20	17	15	0	0
2	10/06/2013	8.45	1025	25%	18	18	1	
3	12/06/2013	1315	1630	100%	15	12	3	Light rain just before and occasional light showers during survey
4	17/06/2013	08:00	19:30	60%	20	16	1	0
5	21/06/2013	0830	1045	15%	16	16	1	none
6	18/07/2013	0545	700	50%	17	17	0	none
7	22/07/2013	1745	1900	45%	17	17	2	heavy showers earlier in day

Assessment

3.6 Reptile populations were assessed in accordance with population level criteria as stated in the Key Reptile Site Register (HGBI, 1998). This system classifies populations of individual reptile species into three population categories assessing the importance of the population (Table 2). These categories are based on the total number of animals observed during individual survey occasions.

Table 2: Key Reptile Site Survey Assessment Categories (HGBI 1998)

Species	Low Population (No. of individuals)	Good Population (No. of individuals)	Exceptional Population (No. of individuals)
Adder	<5	5 - 10	>10

Species	Low Population (No. of individuals)	Good Population (No. of individuals)	Exceptional Population (No. of individuals)
Common lizard	<5	5 - 20	>20
Grass snake	<5	5 - 10	>10
Slow worm	<5	5 - 20	>20

4.0 RESULTS & ASSESSMENT

Desktop study

4.1 Reptile records were provided by the WSBRCt. The general area is known to have supported three species: grass snake *Natrix natrix*, slow worm *Anguis fragilis* and common lizard *Zootoca vivipara.* Only grass snake has been recorded from within the site.

4.2 Grass snake is widely recorded from the local area particularly to the east. Records of the other two species are more restricted with only slow worm being recorded in recent years (2001) in the vicinity of the site. A summary of records is provided in Table 3 below and presented spatially in Figure 1:

Table3: Desk Study reptile data

Name	Location	Souce
Grass snake Natrix natrix	SU1888 ï 1km west of site	WSBRC
Grass snake Natrix natrix	SU190886 ï 800m north of site	WSBRC
Slow worm Anguis fragilis	SU1885 ï 1km west of site	WSBRC
Viviparous lizard Lacerta vivipara	SU205835 ï 700m east of site	WSBRC
Viviparous lizard Lacerta vivipara	SU1885 ï 1km west of site	WSBRC

Field survey

Habitat Assessment

4.3 The habitats within the site comprised a mixture of habitats mainly dominated by agricultuiral management. These included tall grassland cut for hay, large areas of arable formland comprising a muixture of turf and flax, hedgerows and plantation woodland. In addition to the South Marston Brook, which flows through the site, small to medium sized ponds also occur.

4.4 The areas of woodland and tall grassland presented a heavily shaded habitat and or were of a very homogenous vegetation structure that, with the exception of the edge habitat between it and hedgerows, was considered to be largely unsuitable for reptiles, although occasional use could occur during the warmer summer months; by slow worm and grass snake for example. The open arable habitats did not provide any suitable opportunities for reptiles to shelter and were therefore considered unsuitable for this species group. Woodland/hedgerow edge and areas of tipped

material/ruderal vegetation were more suitable; providing the appropriate juxtapostion of areas of shelter, basking and open habitat with a relatively diverse vegetation structure.

Reptile Survey

4.5 Survey of suitable habitat confirmed the presence of grass snake. No other reptile species were recorded during the surveys.

4.6 Grass snakes were recorded mainly in the north of the site in association with the unmanaged grassland. A summary of reptile survey conditions and results is provided in Table 4 below:

Table 4: Reptile survey conditions and results

Survey	Date and Time	Grass snake	Common lizard	Slow worm	Other Species
1	31/05/2013	**1** - rubble and tipped material close to Acorn Bridge	0	0	0
2	10/06/2013	**3** - rubble and tipped material close Acorn Bridge			
3	12/06/2013	**3** - single close to pond at south Marston, single in rubble close to Acorn Bridge, single adjacent to South Marston Brook	0	0	0
4	17/06/2013	**3** ï two juveniles adjacent to pond and South Marston Brook, single adult adjacent to pond south of South Marston	0	0	0
5	21/06/2013	**1**adult - rubble and tipped material close to Acorn Bridge	0	0	0
6	18/07/2013	**1** adult - rubble and tipped material close to Acorn Bridge	0	0	0
7	22/07/2013	None	0	0	0

4.7 The majority of grass snakes (7 of a total of 12) were located in an area of tipped material and rubble close to the River Cole and Acorn Bridge in the extreme east of the survey area. This is commensurate with the wider distribution of grass snake, which has a strong correlation with wetland habitats and river corridors.

4.8 Two further incidental sights were also made, with single a grass snakes observed adjacent to the South Marston Brook and the the south of South Marston, where an adult grass snake was observed dead on

Population Size Assessment

4.9 The results indicate that the site supported a low population (<5 adults) of grass snake, which, given the size of the site, is unremarkable.

4.10 It is likely that the reptiles recorded during survey are likely to form part of a population that occupies suitable habitat in the wider landscape. In a wider context grass snake are relatively

widespread throughout suitable habitat locally. Therefore, it is considered likely that each of the reptile populations are of no more than Local significance.

5.0 RECOMMENDATIONS

5.1 The following section provides details of mitigation required and an assessment of residual impacts on grass snake as a result of the proposals. Implementation of the following mitigation strategies is considered sufficient to ensure that the conservation status of local reptile populations is maintained.

5.2 Given the nature and location of further reptile records it remains a possibility that slow worm and common lizard could occur within the site at a very low density so as to preclude detection by normal methods or alternatively they could occur close to the site, such as along the railway corridor that forms the southern site boundary and where more suitable and extensive habitat occurs. The mitigation measures described here will however ensure that they are not injured during works

Assessment of Potential Impacts and Mitigation Strategies

5.3 The use of habitats by grass snake was scattered (Figure 2). Without the use of mitigation strategies there is potential for an adverse impact on reptile species caused by:

- temporary loss of habitat through vegetation clearance,
- permanent loss of habitat due to the development of hard standing or buildings,
- Incidental harm during earthworks/site clearance.

5.4 Mitigation measures aim to ensure that grass snake is not killed or injured during works and that their local conservation status is maintained.

5.5 The precise method of translocation is greatly dependant on the nature of the construction programme and the phasing of development. Important factors include the extent of working area associated with each phase and the nature of habitat within it. Although governed to a degree by unknown factors and detailed design, it is anticipated that development and site clearance will be phased.

5.6 The highly variable nature of the site, which supports extensive areas of arable farmland that are unlikely to be of significance for the local grass snake population, is likely to necessitate to translocation methods: passive displacement and trapping and translocation. The former will be used where extensive suboptimal habitat occurs whilst the latter is more suitable where open grassland habitat predominates and where grass snake has been more frequently associated.

Receptor Areas

5.7 While passive displacement encourages animals to move freely away from harm to retained habitats outside of the working areas, trapping and translocation requires the active removal of individuals into suitable habitat elsewhere. This will include newly created/enhanced habitats proposed within the GI, such as within the proposed Rowborough or South Marston Parks. The receptor area used will meet a number of criteria, as follows:

- An area that provides suitable habitat
- An area secure from development in the long-term
- An area that maintains the reptiles connectivity with other areas of suitable habitat

5.8 The selected receptor area will be managed prior to trapping and translocation to open up areas and increase the availability of basking habitat. Additional habitat enhancements will include the creation of hibernation features, such as log piles and artificial hibernacula.

5.9 The habitat creation in the receptor area plus additional enhancement measures will ensure that the receptor area provides sufficient carrying capacity to support the translocated reptile populations.

Translocation Methods –Passive Displacement

5.10 Passive displacement will be undertaken where suboptimal habitats occur and the likelihood of reptiles being present greatly reduced. It involves the management of vegetation to displace reptiles towards areas of retained habitat. It is expected that this method would be sufficient for the following:

- Discouraging grass snake from the affected area of work particularly where dese shrub and woodland habitats predominate
- Temporary removal of suitable habitat that is either directly connected to or in close proximity with the areas where reptiles have been recorded

Translocation Proposals –Trapping and Translocation

5.11 Initially, habitats will be managed as above to attempt to encourage reptiles to naturally migrate away from the working corridor. Subsequently the working area will be fenced with reptile exclusion fencing to ensure that removed animals do not return to the affected area. Following this, corrugated tin sheets will be laid within the working areas. Translocation will be undertaken only during the active period Animals captured during the exercise will be recorded and translocated to the receptor areas.

5.12 Once caught, reptiles will be swiftly transferred into clean, cloth bags, taken to the retained habitat and released immediately in order to minimise stress to individuals.

5.13 All potential hibernation sites present within the working area shall be removed carefully by hand by experienced ecologists. This includes piles of rubble and tree roots. Any individual found will be placed in areas of retained habitat or under newly created refugia. New hibernacula in the form of a dry stone wall and hedge bank will be created adjacent to the footpath that bisects the site, to ensure linkage to the existing population along the railway.

5.14 The creation of the new access roads may fragment the existing habitats used by grass snake by forming a barrier to dispersal. The culvert/bridge over the South Marston Brook should be designed so at to minimise its length and maximise its span.

Trapping

5.15 Where appropriate, habitats supporting reptile species will be surrounded by temporary fencing and a high density of artificial refugia which will be used to trap reptiles until a reasonable capture effort has been achieved (English Nature, 2004). The fencing construction would follow the design described in the Design Manual for Roads and Bridges.

5.16 Trapping will make use of artificial refuges, i.e. corrugated tin sheets and felt tiles (approximately $0.5m^2$). In order to maximise the capture rate refugia will be laid at a density of 200 / hectare, which is twice that normally recommended for a population survey (HGBI 1998). Refugia will be laid in an ordered grid pattern throughout the site, according to the prevailing topography and vegetation. The refugia will be given a period of 5 days to 'bed in' before the start of trapping.

5.17 Suitably qualified ecologists will make daily checks of the refugia during suitable weather conditions in the reptile active period (March - October, inclusive), i.e. days with little or no wind when daytime temperature fall between 9 - 18 ^{o}C.

5.18 Any reptiles caught during suitable weather conditions will be hand-caught, recorded and swiftly transferred into buckets or clean, cloth bags. They will then be released directly onto the prepared hibernacula in a receptor site so that stress to the individuals is minimised.

5.19 Vegetation management will also be employed to enhance the rate of capture during the reptile active season. Throughout the trapping period vegetation will be gradually strimmed in order to make it less suitable for reptiles. Strimming shall start from those areas that are already unsuitable for reptiles and move towards the areas of suitable reptile habitat. Examples of unsuitable areas would include those known to be clear of reptiles or grazed areas with a very low sward providing no opportunities for shelter.

5.20 Trapping within the site will continue until a reasonable capture effort has been achieved. This will also be determined by the extent of the area. Translocation will cease when reptiles have been shown to be absent from a phase for a full 5 clear days.

Compensation and Enhancement

5.21 The proposals include extensive areas of retained and / or newly created habitat that will be managed to achieve conservation benefits for the existing reptile populations, along with other species. The management of both receptor areas and other areas of created habitats will be specifically designed to improve both the botanical and structural diversity of vegetation in order to benefit reptiles. These measures will include low intensity management to establish grassland and scrub mosaic, and the provision of a series of additional hibernation features.

5.22 The detailed design of the habitats will be achieved through the implementation of the Mitigation, Habitat and Management Plan, which will ensure the successful establishment and maintenance of all retained and newly created habitats. The management plan will include measures necessary to create and maintain suitable habitats for common reptile species.

5.23 As part of the management plan the population within the site will be monitored. This will include the recording of number and age-classes of reptiles within the receptor site and enhanced habitats. This exercise will be undertaken on three suitable occasions per year for five years following completion of the development.

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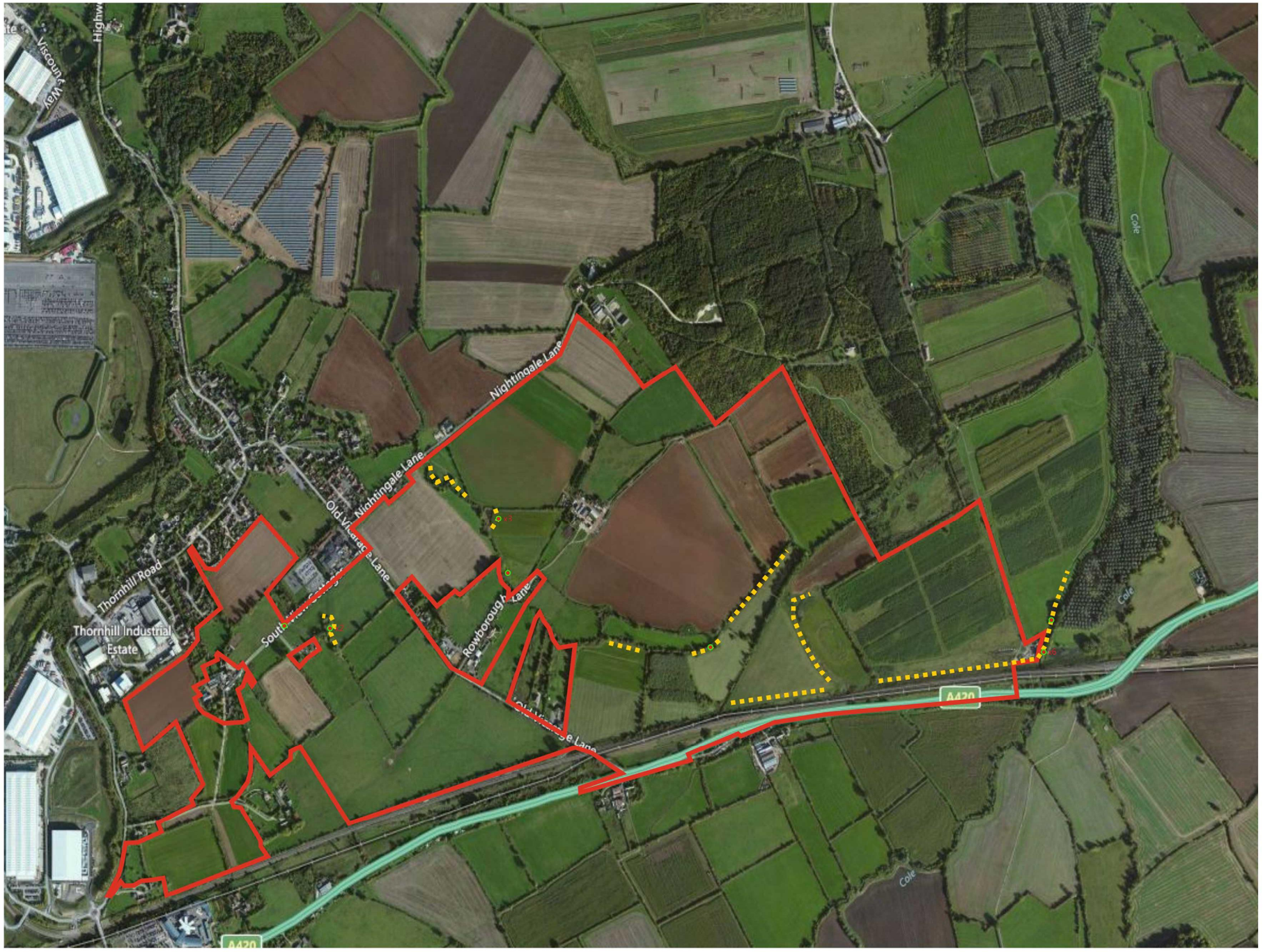

Site Boundary

Refuge locations

Reptile sighting

Hallam Land Management, Taylor Wimpey & Hannick Homes

Eastern Village, Swindon

REPTILE SURVEY PLAN

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Figure 1 **1796-E—**

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